

Inversa unei matrice

Fie $A \in GL_n(K)$, $A = (a_{ij})$, $K \in \{\mathbb{R}, \mathbb{C}\}$

Not cu A^* matricea adjunctă definită prin:

$$A^* = (\Delta_{ij})_{i,j=1}^n, \Delta_{ij} = \text{Complementul algebric al elem } a_{ij}$$

$$\text{Atunci } A^{-1} = \frac{1}{\det A} A^*$$

ex: Det inversele urm matrice (dac există)

$$a) A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 4 & 1 & 4 \end{pmatrix}$$

$$\left| \begin{array}{ccc|c} 1 & 1 & 2 & L_3 - L_1 \\ 2 & -1 & 1 & \\ 4 & 1 & 4 & L_2 + L_1 \end{array} \right| = \left| \begin{array}{ccc|c} 1 & 1 & 2 & \\ 3 & 0 & 3 & \\ 3 & 0 & 2 & \end{array} \right| = - \left| \begin{array}{cc|c} 3 & 3 & \\ 3 & 2 & \end{array} \right| = 3 \neq 0$$

$$\Delta_{11} = (-1)^{1+1} \left| \begin{array}{cc} -1 & 1 \\ 1 & 4 \end{array} \right| = -5$$

$$\Delta_{12} = (-1)^{1+2} \left| \begin{array}{cc} 2 & 1 \\ 4 & 4 \end{array} \right| = -4$$

$$\Delta_{13} = (-1)^{1+3} \left| \begin{array}{cc} 2 & -1 \\ 4 & 1 \end{array} \right| = 6$$

$$\Delta_{21} = -2$$

$$\Delta_{31} = 3$$

$$\Delta_{22} = -4$$

$$\Delta_{32} = 3$$

$$\Delta_{23} = 3$$

$$\Delta_{33} = -3$$

$$A^{-1} = \frac{1}{3} \begin{pmatrix} -5 & -2 & 3 \\ -4 & -4 & 3 \\ 6 & 3 & -3 \end{pmatrix}$$

$$b) \quad B = \begin{pmatrix} 3 & 4 & 2 \\ 6 & 8 & 5 \\ 9 & 12 & 10 \end{pmatrix} \quad \left\{ \Rightarrow \nexists B^{-1} \right.$$

$$\left| \begin{pmatrix} 3 & 4 & 2 \\ 6 & 8 & 5 \\ 9 & 12 & 10 \end{pmatrix} \right| = 0$$

$$c) \quad C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad (\text{terme})$$

Rangul unei matrice

Def = Fie $A \in M_{m,n}(K)$, K corp comutativ, $A \neq O_{m,n}$
Spunem c $\text{rang } A = r$ dac:

- $\exists$ un minor de ordin r al lui A nenul
- toti minorii de ordin $> r$ (dac exista) sunt

$$\text{Definim } \text{rang } O_{m,n} = 0 \quad \left(\binom{r+1}{m} \binom{r+1}{n} + \binom{r+2}{m} \binom{r+2}{n} + \dots + \binom{m}{m} \binom{m}{n} \right) (*)$$

$$\text{Obs: } \text{rang } A \leq \min \{m, n\}$$

$$\bullet \text{rang } A = \text{rang } A^T$$

(*)' : toți minorii de ordin $r+1$ (dacă există) sunt nuli
 $\binom{r+1}{C_m} \binom{r+1}{C_m}$

(*)'' : toți minorii de ordin $r+1$ obținuți prin îndepărtarea lui d (dacă există) sunt nuli

$$((m-r)(n-r) \leq \binom{r+1}{C_m} \binom{r+1}{C_n})$$

ex: determinați rangurile următoarelor matrice

a) $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 \\ 1 & 3 & 0 & -3 \end{pmatrix}$; $\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rang } A \geq 2$

$$\left| \begin{array}{ccc|c} 1 & 1 & 1 & C_2 - C_1 \\ 1 & 0 & 1 & = \\ 1 & 3 & 0 & C_3 - C_1 \end{array} \right| \left| \begin{array}{ccc} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & -3 & 1 \end{array} \right| = (-1)^{3+3} \left| \begin{array}{cc} 1 & 0 \\ 1 & -2 \end{array} \right| = -2 \neq 0$$

$$\Rightarrow \text{rang } A = 3$$

b) $B = \begin{pmatrix} 1 & 2 & 1 & -2 \\ 2 & 3 & 1 & 0 \\ 1 & 2 & 2 & -3 \end{pmatrix}$; $\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = -1 \neq 0 \Rightarrow \text{rang } B \geq 2$

$$\left| \begin{array}{ccc|c} 2 & 1 & -2 & L_3 - L_1 \\ 3 & 1 & 0 & = \\ 2 & 2 & -3 & \end{array} \right| \left| \begin{array}{ccc} 2 & 1 & -2 \\ 3 & 1 & 0 \\ 0 & 1 & 1 \end{array} \right| \xrightarrow{C_3 - C_2} \left| \begin{array}{ccc} 2 & 1 & -3 \\ 3 & 1 & -1 \\ 0 & 1 & 0 \end{array} \right| = 0$$

$$\begin{vmatrix} 2 & -3 \\ 3 & -1 \end{vmatrix} = -2 + 9 = 7 \neq 0 \Rightarrow$$

$$\Rightarrow \text{rang } B = 3$$

$$c) C = \begin{pmatrix} 2 & \alpha & -2 & 2 \\ 4 & -1 & 2\alpha & 5 \\ 2 & 1 & -12 & 1 \end{pmatrix}, \alpha \in \mathbb{C}$$

$$\begin{vmatrix} 2 & 2 \\ 4 & 5 \end{vmatrix} = 2 \neq 0 \Rightarrow \text{rang } C \geq 2$$

$$\begin{vmatrix} 2 & 2 & \alpha \\ 4 & 5 & -1 \\ 2 & 1 & 10 \end{vmatrix} \stackrel{C_2 - C_1}{=} \begin{vmatrix} 2 & 0 & \alpha \\ 4 & 1 & -1 \\ 2 & -1 & 10 \end{vmatrix} \stackrel{L_3 + L_2}{=} \begin{vmatrix} 2 & 0 & \alpha \\ 4 & 1 & -1 \\ 6 & 0 & 9 \end{vmatrix} =$$

$$= \begin{vmatrix} 2 & \alpha \\ 6 & 9 \end{vmatrix} = 18 - 6\alpha = 6(3 - \alpha)$$

Dann $\alpha \neq 3, \alpha \in \mathbb{C} : \text{rang } C = 3$

Dann $\alpha = 3 : \text{rang } C = 2$

$$\begin{vmatrix} 2 & 2 & -2 \\ 4 & 5 & 2\alpha \\ 2 & 1 & -12 \end{vmatrix} \stackrel{C_2 - C_1}{=} \begin{vmatrix} 2 & 0 & -2 \\ 4 & 1 & 2\alpha \\ 2 & -1 & -12 \end{vmatrix} \stackrel{L_3 - L_1}{=} \begin{vmatrix} 2 & 0 & -2 \\ 4 & 1 & 2\alpha \\ 0 & -1 & -10 \end{vmatrix} \stackrel{C_3 + C_1}{=} \begin{vmatrix} 2 & 0 & -2 \\ 4 & 1 & 2\alpha \\ 0 & -1 & -10 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 0 & 0 \\ 4 & 1 & 2\alpha + 4 \\ 0 & -1 & -10 \end{vmatrix} = \begin{vmatrix} 1 & 2\alpha + 4 \\ -1 & -10 \end{vmatrix} = -10 - 2\alpha - 4 =$$

$$= -2\alpha - 14 =$$

$$= -2(\alpha + 7) = 0$$

$$d) \Delta = \begin{pmatrix} 2 & 1 & 1 & 3 \\ 3 & 2 & -1 & 4 \\ \alpha & 3 & 5 & -3 \\ 7 & -5 & 3 & 1 \end{pmatrix}, \alpha \in \mathbb{R}$$

e) rangul matricei Vandermonde (term)

$$d) \begin{vmatrix} 1 & 3 \\ -1 & 4 \end{vmatrix} = 7 \neq 0 \Rightarrow \text{rang } \Delta \geq 2$$

$$\begin{vmatrix} 1 & 3 & 2 \\ -1 & 4 & 3 \\ 3 & 1 & 7 \end{vmatrix} = 28 - 2 + 27 - 24 - 3 + 21 = 47 \neq 0 \Rightarrow \text{rang } \Delta \geq 3$$

$$\begin{vmatrix} 2 & 1 & 1 & 3 \\ 3 & 2 & -1 & 4 \\ \alpha & 3 & 5 & -3 \\ 7 & -5 & 3 & 1 \end{vmatrix} \begin{matrix} C_1 - 2C_3 \\ \\ C_2 - C_3 \\ C_4 - 3C_3 \end{matrix} \begin{vmatrix} 0 & 0 & 1 & 0 \\ 5 & 3 & -1 & 7 \\ 2-10 & -2 & 5 & -18 \\ 1 & -8 & 3 & -8 \end{vmatrix} =$$

$$= \begin{vmatrix} 5 & 3 & 7 \\ \alpha-10 & -2 & -18 \\ 1 & -8 & -8 \end{vmatrix} \begin{matrix} C_3 - C_2 \\ \\ \\ \end{matrix} \begin{vmatrix} 5 & 3 & 4 \\ \alpha-10 & -2 & -16 \\ 1 & -8 & 0 \end{vmatrix} \begin{matrix} L_2 + 4L_1 \\ \\ \\ \end{matrix}$$

$$= \begin{vmatrix} 5 & 3 & 4 \\ \alpha+10 & 10 & 0 \\ 1 & -8 & 0 \end{vmatrix} = 4 \begin{vmatrix} \alpha+10 & 10 \\ 1 & -8 \end{vmatrix} = 4(-8\alpha - 80 - 10) = 4(-8\alpha - 90) = -8(4\alpha + 45)$$

$$\text{Dacă } 8\alpha = 90 \Rightarrow \alpha = \frac{90}{8} = \frac{45}{4} \Rightarrow \text{rang } \Delta = 3$$

$$\text{Dacă } \alpha \neq \frac{45}{4} \Rightarrow \text{rang } \Delta = 4$$

Sisteme de ecuatii liniare

$$(S) \begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases} \Leftrightarrow$$

$$AX = b, \quad A = (A_{ij}) \in M_{m,n}(K)$$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \text{ necunoscute, } b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \text{ term liberi}$$

$$\bar{A} = (A|b) = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{pmatrix} \in M_{m,n+1}(K)$$

matricea
extinsă

Cozul $m=n$: Dacă $\text{rang } A = n \Leftrightarrow \det A \neq 0 \Rightarrow$
 $\Rightarrow$ compatibil determinat (soluție unică)

Soluțiile sunt date de regula lui Cramer:

$$x_1 = \frac{\Delta_1}{\Delta}, \quad x_2 = \frac{\Delta_2}{\Delta}, \quad \dots, \quad x_n = \frac{\Delta_n}{\Delta}$$

unde $\Delta_i = \det$ matricei obținute din A prin
 înlocuirea coloanei i cu cel term liberi

Teoremă: S este compatibil $\Leftrightarrow \text{rang } A = \text{rang } \bar{A}$

Reamintim: Dacă $\text{rang } A = r$ atunci există un minor de ordin r nenul.

Un astfel de minor s.m. minor principal.

Fie Δ un minor principal. Minorii de ordin $r+1$ ai lui $\bar{A}$ obținem bătându-lui Δ cu cel term liber și cu câte o linie care nu intervine în Δ (dacă există) s.m. minorii caracteristici.

Teoremă: (S) compatibil $\Leftrightarrow$ toți minorii caracteristici (comp. unui minor principal (dacă există)) sunt nuli

$$\begin{cases} x_1 + x_2 + 2x_3 = -1 \\ 2x_1 + x_2 + 2x_3 = -4 \\ 4x_1 + x_2 + 4x_3 = -2 \end{cases}$$

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{pmatrix}$$

$$\bar{A} = \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 2 & -1 & 2 & -4 \\ 4 & 1 & 4 & -2 \end{array} \right)$$

$$\det A = \begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{vmatrix} = -4 + 4 + 8 + 8 - 2 - 8 = 6 \neq 0 \Rightarrow$$

~~rang A~~ (S) comp det

(Cramer)